

Reconnaissance & Vulnerability Scanning Report

1. Objective

The objective of this task was to perform active reconnaissance and vulnerability scanning on safe test systems using cybersecurity tools such as Nmap and Nikto. The assessment focused on identifying open ports, running services, server configurations, and potential security vulnerabilities.

2. Tools Used

Tool	Purpose
Nmap	Network reconnaissance and service enumeration
Nikto	Web server vulnerability scanning
Kali Linux	Penetration testing operating system

3. Target Information

Target 1

- Hostname: scanme.nmap.org
- IP Address: 45.33.32.156

Target 2

- Hostname: testaspnet.vulnweb.com
 - IP Address: 44.238.29.244
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4. Nmap Reconnaissance Scan

Command Used

nmap -sV -A scanme.nmap.org

Purpose of the Command

- **-sV → Detects service versions**
- **-A → Enables OS detection, version detection, script scanning, and traceroute**

5. Nmap Scan Results

The scan successfully identified several open and filtered ports on the target system.

Open Ports and Services

Port	State	Service	Version
22/tcp	Open	SSH	OpenSSH 6.6.1p1 Ubuntu
80/tcp	Open	HTTP	Apache httpd 2.4.7
9929/tcp	Open	Nping Echo	Nping Echo
31337/tcp	Open	tcpwrapped	Unknown

Filtered Ports

Port	Service
25/tcp	SMTP
135/tcp	MSRPC
139/tcp	NetBIOS-SSN
445/tcp	Microsoft-DS

6. Additional Nmap Findings

- Operating System detected as Linux Kernel 5.x
 - Apache Web Server running on Ubuntu
 - SSH service exposed publicly
 - HTTP service accessible on port 80
 - Host latency detected at 0.27 seconds
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7. Security Risks Identified from Nmap

Issue	Risk Level	Description
Open SSH Port	Medium	SSH service may be targeted for brute-force attacks
Apache 2.4.7 Version	Medium	Older versions may contain vulnerabilities
Public HTTP Service	Medium	Web services can expose sensitive information
Multiple Open Services	Medium	Increased attack surface

8. Screenshot Evidence (Nmap)

Screenshot – Nmap Scan Results



9. Nikto Vulnerability Scan

Command Used

nikto -h http://testaspnet.vulnweb.com

10. Nikto Scan Findings

Nikto identified multiple security misconfigurations and potential vulnerabilities on the target web server.

Server Information

Item	Details
Web Server	Microsoft-IIS/8.5
ASP.NET Version	2.0.50727
Framework	ASP.NET

11. Vulnerabilities Identified

Vulnerability	Risk Level	Description
Missing X-Frame-Options Header	Medium	Could allow clickjacking attacks
Missing X-Content-Type-Options Header	Medium	Browser may incorrectly interpret content types
Internal IP Disclosure	Medium	Internal IP address "10.0.0.14" exposed
HTTP TRACE Method Enabled	Medium	TRACE method may allow Cross Site Tracing attacks
Public HTTP Methods Enabled	Low	Multiple HTTP methods exposed
test.txt File Found	Low	Potentially sensitive file exposed

12. Important Nikto Findings

Missing Security Headers

The target server did not implement important HTTP security headers:

- X-Frame-Options
- X-Content-Type-Options

This may expose users to clickjacking and MIME-sniffing attacks.

Internal IP Address Disclosure

Nikto detected disclosure of a private internal IP address:

- 10.0.0.14

This information could assist attackers during internal network reconnaissance.

HTTP TRACE Method Enabled

The HTTP TRACE method was enabled on the server, which can increase the risk of Cross-Site Tracing (XST) attacks.

13. Screenshot Evidence (Nikto)

Screenshot – Nikto Scan Results

```
(abhisith@kali) ~$ nikto -h http://testaspnet.vulnweb.com
- Nikto v2.5.0

+ Multiple IPs found: 44.238.29.244, 64:ff9b::2cee:1df4
+ Target IP: 44.238.29.244
+ Target Hostname: testaspnet.vulnweb.com
+ Target Port: 80
+ Start Time: 2026-06-03 21:18:47 (GMT5.5)

+ Server: Microsoft-IIS/8.5
+ /: Retrieved x-aspnet-version header: 2.0.50727.
+ /: Retrieved x-powered-by header: ASP.NET.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type.
  See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ RFC-1918 /aspnet_client: IP address found in the 'location' header. The IP is "10.0.0.14". See: https://portswigger.net/kb/issues/00600300_private-ip-addresses-disclosed
+ /aspnet_client: The web server may reveal its internal or real IP in the Location header via a request to with HTTP/1.0. The value is "10.0.0.14". See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2000-0649
+ OPTIONS: Allowed HTTP Methods: OPTIONS, TRACE, GET, HEAD, POST .
+ OPTIONS: Public HTTP Methods: OPTIONS, TRACE, GET, HEAD, POST .
+ /test.txt: This might be interesting.
+ ERROR: Error limit (20) reached for host, giving up. Last error: opening stream: can't connect (connect error): Network is unreachable
+ Scan terminated: 12 error(s) and 9 item(s) reported on remote host
+ End Time: 2026-06-03 21:45:22 (GMT5.5) (1595 seconds)

+ 1 host(s) tested
```

14. Recommendations

The following recommendations are suggested to improve the security posture of the target systems:

- 1. Disable unused open ports and services**
 - 2. Update outdated software versions**
 - 3. Disable HTTP TRACE method**
 - 4. Implement missing HTTP security headers**
 - 5. Hide internal IP addresses from responses**
 - 6. Regularly perform vulnerability assessments**
 - 7. Use firewall rules to restrict unnecessary access**
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15. Conclusion

This task provided practical experience in reconnaissance and vulnerability assessment using Nmap and Nikto. The scans successfully identified open services, server details, and multiple security weaknesses. Proper vulnerability management and regular security testing are important to reduce potential attack surfaces and improve system security.

16. References

- Nmap Official Documentation**
- Nikto Official Documentation**
- OWASP Security Guidelines**